



IIT JODHPUR

Department Of CSE

Bayesian Networks for Complex Decision-Making in Healthcare Management

CSL 7620: Machine Learning

Project Report

Submitted By:

Team 12

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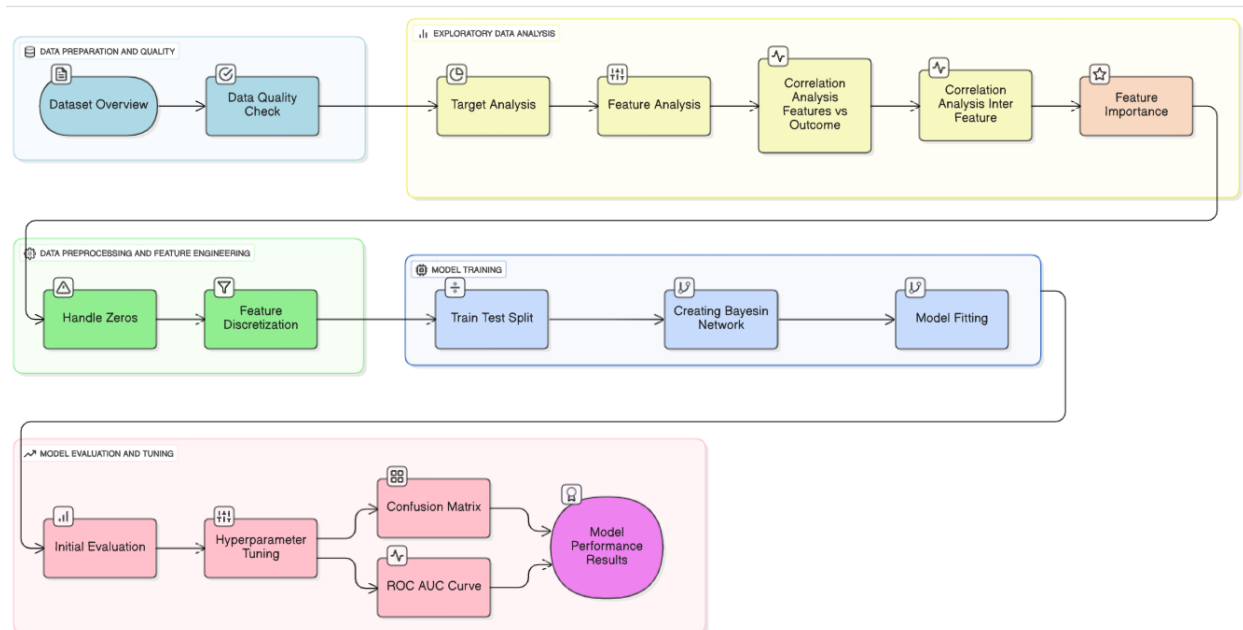
1 . Streamlit link:

We have deployed our model in Github using Streamlit.I am attaching the link for the same.

<https://diabetes-predictor-5rjtswvkdvrvdscxtpon63.streamlit.app/#prediction-results>

Note: We have taken help of LLM for deploying the model to Streamlit.

2 .Project Architecture:



3 .Dataset Overview :

- Total Records: 768 patients
- Total Features: 9 attributes
- Target Variable: Outcome (Binary: 0 or 1)
- Data Quality: Complete records with zero missing values
- Data Type: all numerical continuous values

4 .Model Training Methodology:

1. Scoring Method: Bayesian Information Criterion (BIC)

The BIC score is used to evaluate different network structures. BIC balances model fit with complexity, penalizing overly complex networks to prevent overfitting. Lower BIC scores indicate better models.

2. Incorporating Expert Knowledge

To ensure clinical validity, two types of constraints are imposed on the structure learning process:

Blacklist (Forbidden Edges): These edges represent relationships that are medically implausible or causally impossible:

- BMI → Age: BMI cannot cause age
- Pregnancies → Age: Number of pregnancies cannot cause age
- SkinThickness → BMI: Skin thickness doesn't cause BMI changes

Whitelist (Required Edges): These edges represent well-established medical relationships that must appear in the network:

- Glucose → Outcome: High blood glucose is a primary diabetes indicator
- BMI → Outcome: Obesity is a known diabetes risk factor
- Pregnancies → Outcome: Gestational diabetes increases risk
- BloodPressure → Outcome: Hypertension is linked to diabetes
- Age → Pregnancies: Age influences reproductive history
- BMI → SkinThickness: Body mass correlates with skin fold thickness

3. Hill Climbing Search Algorithm

The structure learning uses the **Hill Climbing Search** algorithm with the **Expert knowledge**

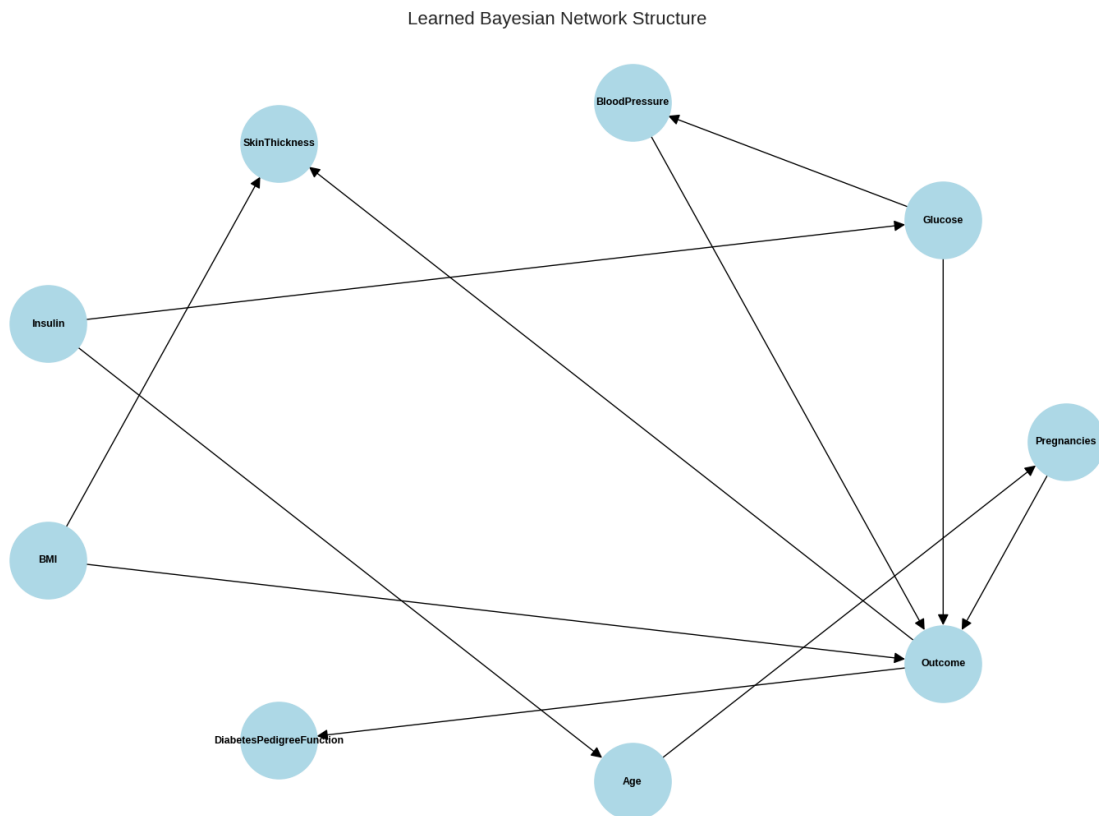
The algorithm iteratively modifies the network structure (adding, removing, or reversing edges) and evaluates each change using the BIC score, ultimately converging on the structure with the best score while respecting medical constraints.

4. Advantages of This Approach

1. **Clinical validity:** Expert constraints ensure medically meaningful relationships
2. **Data-driven optimization:** Structure learning discovers patterns not obvious a priori
3. **Interpretability:** The resulting network shows clear causal pathways

4. **Uncertainty handling:** Bayesian approach naturally models probabilistic relationships
5. **Balanced complexity:** BIC scoring prevents both underfitting and overfitting

Visualization



5 .Baseline Results Comparison

Model Performance Summary:

Model Performance Comparison:			
	ROC-AUC	F1-Score	Accuracy
Random Forest	0.945926	0.844037	0.889610
SVM	0.905185	0.800000	0.850649
Bayesian Network	0.843241	0.730435	0.798701
Logistic Regression	0.856481	0.660000	0.779221

Key Finding: **BAYESIAN NETWORK** beats Logistic Regression in Accuracy but not able to beat Random Forest and SVM.

Optimization Method: SimpleGridSearch

Performance Improvement:

- Accuracy improvement: +1.3% over baseline
- F1- score Improvement: +0.35
- Generalization: Better performance on test set

6 .Conclusion

- **Computational Complexity:** Bayesian networks require significant computational resources for inference and learning, especially as the network structure becomes more complex, making them slower in practice.
- **Scalability Issues:** They struggle with high-dimensional datasets that have many variables, while Random Forests and SVMs handle large feature spaces more effectively through parallelization.
- **Discretization Requirements:** Bayesian networks often need continuous variables to be discretized into bins, which leads to information loss and reduced predictive accuracy.
- **Limited Non-linearity Capture:** They may not capture complex non-linear relationships as effectively as SVMs with kernel tricks or Random Forests with their ensemble of decision trees.
- **Training Time:** Learning the optimal network structure and parameters from data is computationally expensive and time-consuming compared to training Random Forests or SVMs.
- **Black Box Trade-off:** While Bayesian networks offer interpretability, this comes at the cost of predictive performance, whereas Random Forests and SVMs prioritize accuracy even if they sacrifice some explainability.
- **Sensitivity to Structure:** The performance heavily depends on getting the network structure right, and incorrect structure specification can significantly degrade results compared to the more robust nature of ensemble methods.